**IT4010 Research Project – 2025 July**

**Topic Assessment Form**

**Project ID :** 25-26J-201

1. Topic (12 words max)

Integrated Home Design and Planning Management System

2. Research group the project belongs to

**CoEAI - Centre of Excellence for AI**

3. Specialization of the project belongs to

**Information Technology (IT)**

4. If a continuation of a previous project:

| | |
|---|---|
| Project ID | |
| Year | |

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Creating a home is one of the most significant investments for individuals and families, particularly in developing countries such as Sri Lanka, where owning a house is often considered a lifelong achievement. However, the process of planning and building a house is highly complex and presents multiple challenges for prospective homeowners. The increasing demand for housing, especially in urban areas, has led to issues such as limited space, inefficient designs, and a lack of proper alignment with cultural practices and environmental needs. The core challenge lies in integrating modern design techniques with traditional cultural values while ensuring that the design is efficient, functional, and sustainable. This has been further exacerbated by the recent economic crisis, which has led to a surge in construction costs and a decline in new projects [4].

One of the primary issues homeowners face is interior space optimization. In many cases, traditional methods of space planning do not take full advantage of the available area, leading to wasted space and inefficient layouts. The current approach to home design is often fragmented, requiring homeowners to engage with multiple professionals—such as architects, engineers, and contractors—and to manually coordinate each stage of the project. This siloed process increases costs, causes delays, and significantly raises the risk of miscommunication and errors [2]. Without a centralized, integrated platform, it is difficult for homeowners to visualize their project holistically, from initial concept to final construction plan, hindering their ability to make informed decisions.

A second significant problem is the disconnection between modern design and traditional knowledge, specifically principles like Vastu Shastra. These principles, which are deeply rooted in

armony between a structure and its natural environment, optimizing elements such as natural light, ventilation, and energy flow [6]. However, modern, generic designs often overlook these crucial cultural and spiritual aspects. This creates a dilemma for homeowners who want a contemporary home that also respects their heritage. There is a need for a solution that can intelligently incorporate these principles into a modern design framework, allowing for homes that are both aesthetically pleasing and culturally resonant.

Finally, the construction industry in Sri Lanka faces a critical need to shift towards sustainable practices. The current over-reliance on non-renewable, high-cost imported materials not only drives up construction costs but also contributes to a negative environmental impact [7]. There is a clear need to explore and integrate traditional, local, and sustainable building materials into modern designs to reduce a project's carbon footprint and overall cost [3]. This research project aims to address these multifaceted challenges by developing an integrated, intelligent system that streamlines the home design process, respects cultural heritage, and promotes sustainable construction, thereby making the dream of homeownership more accessible and environmentally conscious for all Sri Lankans [8].

**References**

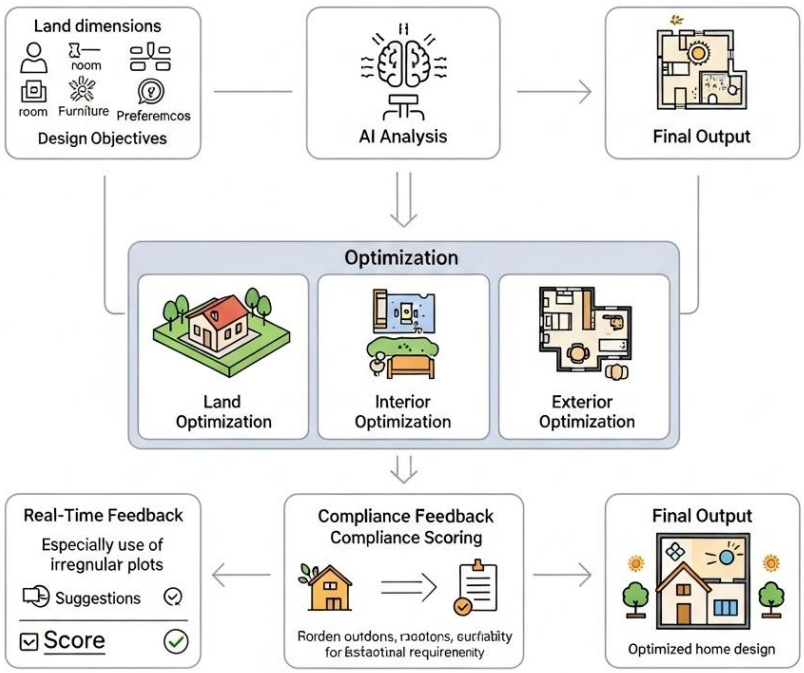
1. M. A. Ranwatta, “A study of the impact of minimalism on user psychology,” Department of Architecture, University of Moratuwa, Moratuwa, Sri Lanka, 2021.
2. H. H. Ekanayake, et al., “Modelling design stage risks in modular integrated construction projects in Sri Lanka,” *ResearchGate*, 2023. Available: https://www.google.com/search?q=https://www.researchgate.net/publication/372830267.
3. K. B. C. Fernando and A. Liyanage, "Exploring traditional building materials for urban housing in Sri Lanka," *Journal of Tropical Architecture*, 2023. Available: https://journals.sjp.ac.lk/index.php/icremv/article/download/6773/4805/18541.
4. T. Hettiarachchi and M. Dhanji, "The Challenges of Sustainable Affordable Housing Construction During the Sri Lankan Economic Crisis," *International Journal of Research and Innovation in Social Science*, vol. 8, no. 2, pp. 1868-1875, 2024.
5. D. S. D. W. K. Liyanagamage, et al., "Survival Challenges Faced by the Small-scale Building Contractors in Sri Lanka," *Proceedings of the 12th World Construction Symposium*, pp. 415-424, 2024.
6. A. G. B. G. L. Herath, et al., "A study on the integration of Vaastu Shastra principles with sustainable building practices," *Journal of Multi-disciplinary Research*, vol. 2, no. 3, 2023.
7. M. K. Dissanayake, et al., "Towards a sustainable building stock in Sri Lanka," *IESL-Sri Lanka Engineering News*, vol. 53, no. 02, pp. 4-6, 2022.
8. R. A. S. Ranathunga and E. P. S. K. Ekanayake, "An investigation into challenges and mitigation measures for implementing green building practices in Sri Lanka," *International Journal of Environmental and Sustainability Engineering*, vol. 1, no. 2, pp.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The solution is an AI-powered platform designed to optimize residential home designs by improving both the functionality and efficiency of space while integrating Vastu Vidya principles for cultural alignment. The system begins by taking user inputs, including land dimensions and specific requirements like the number of rooms and outdoor space. Once this information is provided, the AI analyzes the layout and generates space optimization suggestions, ensuring efficient room usage and traffic flow.

An additional feature allows users to align their design with Vastu Vidya, adjusting room orientations, furniture placements, and overall layout to promote spiritual harmony. For users working with smaller plots of land, the system offers recommendations for land optimization, ensuring the most efficient use of the available space for both the house and garden.

Furthermore, the system addresses exterior space optimization by maximizing the functionality of gardens, rooftops, and facades. Whether for relaxation, gardening, or energy-efficient designs, the system ensures these outdoor areas serve multiple purposes. It also provides real-time feedback and generates a compliance score, helping users see how well their designs adhere to both Vastu Vidya principles and space optimization goals. This comprehensive tool ensures that homes are not only efficient and functional but also culturally aligned and spiritually balanced.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

To successfully develop the AI-powered Integrated Home Design and Planning Management System, specialized knowledge and expertise are required in several areas, including artificial intelligence, architectural design, and cultural principles like Vastu Vidya.

1. **Artificial Intelligence (AI) and Machine Learning:** The system will rely heavily on AI and machine learning to analyze floor plans, optimize space usage, and provide real-time feedback. Expertise in training AI models to detect inefficiencies in layout and suggest real-time improvements is essential. Additionally, AI will be used to align designs with Vastu Vidya principles, requiring knowledge of rule-based AI to integrate cultural guidelines into the system.
2. **Architectural Design and Space Optimization:** Understanding space planning and ergonomic design principles is crucial. The AI system must analyze room dimensions and efficiently place furniture while considering factors like traffic flow, comfort, and space usage. Expertise in interior design and land optimization will help the system generate practical, efficient layouts, particularly for smaller plots.

3. **Vastu Vidya:** Vastu Vidya is the traditional architectural system based on spatial alignment for promoting harmony. A deep understanding of Vastu principles such as room orientation, furniture placement, and environmental alignment is needed to incorporate this into the design process.
This knowledge will guide the AI in making design decisions that respect cultural practices while ensuring functionality.

4. **Data Requirements:** The system will need several types of data:

- **Floor plan data**: Dimensions of rooms and furniture placements.
- **User preferences**: Specific requirements for room functions, furniture types, and aesthetic choices.
- **Vastu Vidya guidelines**: Cultural data on room placements and furniture orientations.
- **Land dimensions**: Data on land plots for land optimization.

8. Objectives and Novelty

| Main Objective<br>The primary objective of this research is to develop an AI-powered platform that integrates Vastu Vidya-based land optimization, real-time design corrections, and interior space optimization. The system will optimize land use, ensure cultural alignment through Vastu principles, and improve interior functionality and space efficiency. It will also optimize exterior spaces such as gardens and rooftops. The platform provides a compliance score and real-time feedback to ensure designs are both practically efficient and spiritually aligned, offering a holistic solution to modern home design challenges. | | | |
|---|---|---|---|
| Member Name with Registration No | Sub Objective | Tasks | Novelty |
| Rathnasinghe S.J.R<br>IT22908124 | Develop an AI-powered platform that optimizes interior space in residential homes, ensuring maximum efficiency and comfort. The system will generate optimized furniture placements and layouts for all rooms, improving overall space utilization while also integrating | **Interior Space Optimization**:<br>The system will analyze room dimensions and furniture types to provide suggestions for optimal placement, ensuring that spaces are not overcrowded and that traffic flow, ergonomics, and user comfort are maximized. In addition, the system will integrate Vastu Vidya principles, guiding furniture placement to ensure alignment with traditional cultural guidelines. For example, it will suggest placing the dining table facing east, a practice recommended in Vastu Vidya, alongside optimizing the room layout for better space use. | **Real-time Design Adjustments & Feedback**:<br>The platform will offer instant corrections based on user input, adjusting furniture placement and room layout for better space utilization. It will also provide real-time feedback, helping users visualize the changes and understand the impact on space efficiency. Additionally, the system will incorporate Vastu Vidya principles in real-time, offering users the option to align their home designs with both functional and cultural requirements. |

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| | Vastu Vidya principles for room orientation and furniture placement. | | |
| Dinetha K A V<br>IT22924278 | Develop an AI-powered system that ensures residential floor plans are in harmony with Vastu Vidya principles, seamlessly integrating cultural, spiritual, and environmental standards into the design process. The system will also generate 2D floor plans based on the specific characteristics of the residential design. | **Vastu Vidya & Design Correction**<br>• The platform will automatically analyze floor plans for Vastu Vidya compliance, making real-time corrections to elements such as room orientation, furniture placement, and overall layout. It will provide a compliance score, offering insights into how well the design adheres to traditional principles of Vastu Vidya. | **Real-time Design Adjustments & Automated Compliance Scoring**<br>The system offers real-time design adjustments, providing instant feedback and corrections, unlike the traditional time-consuming process of manual reviews. It also automatically generates a compliance score, making it faster and more efficient to ensure Vastu alignment. |

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| Sunkalpani S M D R IT22914996 | Develop an AI-based system for land optimization and plan generation that ensures efficient use of available land while maintaining cultural alignment with Vastu Vidya principles. The system will optimize plot layouts and suggest where the house, garden, and other features should be placed. | **Land Optimization & Plan Generation**: The system will analyze the dimensions of the land and user requirements to suggest the most efficient layout. It will also align the plan with Vastu Vidya, ensuring the house placement adheres to cultural principles like room orientation and the placement of key areas such as the kitchen, living room, and bedrooms. | **Compliance with Vastu Vidya & Real-time Feedback**: By integrating Vastu Vidya, the system will ensure the layout is both efficient and culturally aligned. It will provide real-time design adjustments, making the layout process faster and more efficient, while offering insights through compliance scores. |

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| Hansika J A J<br>IT22346018 | Design an AI-powered platform that focuses on optimizing the use of exterior spaces such as gardens, rooftops, and facades for functional purposes like gardening, relaxation, and energy efficiency. The system will ensure these areas are optimized for both aesthetic and practical use. | **Exterior Space Optimization**:<br>The platform will analyze the dimensions of outdoor spaces and suggest the best use for gardens, rooftops, and facades, taking into account the available space and user preferences. It will provide options for sustainable and energy-efficient designs like natural cooling and green spaces. | **Energy-efficient Designs & Real-time Adjustments**:<br>The system will suggest optimized designs for exterior areas to improve energy efficiency and overall usability. It will also provide real-time feedback on design choices, ensuring that the outdoor spaces are as functional and sustainable as possible. |

9. Individual component description of how it is complied with the specialization.

| Member Name with Registration No | Description |
|---|---|
| Rathnasinghe S.J.R<br>IT22908124 | **Interior Space Optimization:** The AI-powered system for interior space optimization leverages machine learning algorithms to analyze room dimensions and furniture types to suggest the best placement. This ensures the maximum efficiency of the living spaces by improving traffic flow, ergonomics, and overall comfort. The system also integrates Vastu Vidya principles to guide furniture placement, such as the dining table orientation, ensuring the design is both functional and spiritually aligned with traditional cultural practices. This integration of space optimization and Vastu Vidya principles provides a holistic solution for optimizing interiors while respecting cultural heritage. |
| Dinetha K A V<br>IT22924278 | **Vastu Vidya & Design Correction:** This module ensures residential floor plans align with Vastu Vidya principles, integrating cultural, spiritual, and environmental standards. It automatically analyzes designs for compliance, making real-time adjustments to room orientation, furniture placement, and layout. The system provides a compliance score, offering quick, automated feedback and corrections, streamlining the traditional, time-consuming review process for faster Vastu alignment. |
| Sunkalpani S M D R<br>IT22914996 | **Land Optimization & Plan Generation:** The AI system for land optimization and plan generation analyzes the dimensions of the land and the user's requirements to recommend the most efficient layout. The system optimizes the placement of the house, garden, and other features within the available plot. By integrating Vastu Vidya, the layout is also adjusted to respect traditional principles, such as room orientation and the proper placement of key areas like the kitchen and living spaces. This ensures that the layout is both spatially efficient and culturally aligned, creating a balance between modern functionality and traditional values. |
| Hansika J A J<br>IT22346018 | **Exterior Space Optimization:** The AI-powered system for exterior space optimization focuses on maximizing the use of outdoor areas, such as gardens, rooftops, and facades, for both aesthetic appeal and functionality. It analyzes available space and recommends layouts that improve comfort, energy efficiency, and sustainability. The system also ensures that exterior spaces are designed for recreation, gardening, and natural cooling, aligning with cultural expectations and promoting green living. By leveraging AI, this component maximizes outdoor space potential, enhancing the living environment while addressing environmental concerns. |

10. Supervisor details

| | Title | First Name | Last Name | Signature |
|---|---|---|---|---|
| Supervisor | | | | |
| Co-Supervisor | | | | |
| External Supervisor | | | | |
| Summary of external supervisor's (if any) experience and expertise | | | | |

**This part is to be filled by the Topic Screening Staff members.**

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

| Yes | | No | |
|---|---|---|---|

b) Does the proposed topic exhibit novelty?

| Yes | | No | |
|---|---|---|---|

c) Do you believe they have the capability to successfully execute the proposed project?

| Yes | | No | |
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d) Do the proposed sub-objectives reflect the students' areas of specialization?

| Yes | | No | |
|---|---|---|---|

e) Supervisor's Evaluation and Recommendation for the Research topic:

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| IT4010 Research Project – 2025 July<br><br>Topic Assessment Form |
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Acceptable: Mark/Select as necessary

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|---|---|
| Topic Assessment Accepted | |
| Topic Assessment Accepted with minor changes* | |
| Topic Assessment to be Resubmitted with major changes* | |
| Topic Assessment Rejected. Topic must be changed | |

* Detailed comments given below

Comments

| |
|---|
| |

| Staff Member's Name | Signature |
|---|---|
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*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.